

SIMATS ENGINEERING ASSESSMENT TOOL 3

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA1703

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. (BL3)

Assessment Tool Weightage: 10%

QUESTIONS: 2

Duration : 45 Minutes
Total Marks:20

Annexure A

Descriptive Written Test

Code of Conduct:

I B. Siva Sankar Reddy ⁽¹⁹²⁴⁷²²⁷¹⁾ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

B. Siva Sankar Reddy

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✓ 20

Annexure A

Descriptive Written Test

1. A smart home automation system is being designed to manage lighting, temperature, and security based on user behavior and environmental conditions. The system must respond to real-time inputs such as motion detection, temperature changes, and time of day while also learning user preferences over time. In this context, explain the different types of intelligent agents (simple reflex, model-based, goal-based, and utility-based agents) and analyze how each type would function within this system. Justify which type of agent or hybrid approach would be most effective for ensuring comfort, energy efficiency, and security.
2. A robot is placed in an unknown maze and must find a path to the exit without prior knowledge of the layout. Explain how uninformed search strategies like Uniform Cost Search (UCS) and Iterative Deepening Search (IDS) can help solve this problem.
Discuss the advantages and disadvantages of these algorithms in terms of time and space complexity. Relate the problem to different types of intelligent agents and explain how agent design affects decision-making. Suggest which combination of agent type and search strategy would be most effective and justify your choice.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Criteria	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Max Marks
Understanding of Concepts	25%	Demonstrates thorough, accurate understanding of concepts	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor understanding; major misconceptions
Explanation	25%	Detailed, well-explained answers with relevant examples	Clear explanation with adequate details	Limited explanation; lacks depth	Poor or unclear explanation
Application	20%	Effectively applies concepts with strong analytical ability	Applies concepts with minor errors	Limited application with weak analysis	No application or incorrect analysis
Organization	15%	Well-structured answers with logical flow and coherence	Organized with minor issues in flow	Basic structure, lacks clarity	Poor organization, incoherent answers
Accuracy	10%	Completely accurate and covers all required points	Mostly accurate with minor omissions	Partially correct with missing points	Incorrect answers with major gaps
Clarity	5%	Clear, neat, professional presentation	Understandable with minor issues	Limited clarity, readability issues	Poorly written, difficult to understand

Intelligent Agents in Smart Home Automation

A smart home automation system must respond to real-time inputs (motion, temperature, time) and learn user preferences. Different types of intelligent agents can be applied as follows:

1) Simple Reflex Agent

operates based on current percepts only (Condition-action rules)

Example:

- If motion detected $\rightarrow$ Turn on lights
- If temperature $>$ threshold $\rightarrow$ Turn on AC
- Limitation: Cannot learn or adapt; works only for predefined conditions.

2) Model-Based Agents

- Maintains an internal state (model of environment)
- Uses past data + current input

Ex

- Remembers room occupancy patterns.
- Adjusts lighting even if motion sensor briefly fails.
- Advantage: Handles partially observable environments better.

3) Goal-Based Agent

- Acts to achieve specific goals

Ex Goals:

Maintain room temperature at 24°C

- Ensure house security

- Evaluates different actions to reach the goal

Advantage: More flexible decision Making.

1) Utility-Based Agent

- Chooses actions Based on maximum utility.
- Balances multiple objectives such as comfort, Energy efficiency and security.

Ex:- If the room is slightly warm

Comparison

Agent type	memory	Goal	Learning
Simple - Reflex	No	No	No
Model-Based	Yes	No	Limited
Goal-Based	Yes	Yes	No
Utility-Based	Yes	Yes	Yes.

Effective Choice

- Learn user preferences over time
- Maintain comfort by adjusting lights and temperature.

Conclusion

A hybrid Intelligent agent provides the best solution for smart home automation because it combines memory, planning and optimization. It ensures comfort energy efficiency and security making the home more intelligent and user friendly.

Introduction

A robot placed in an unknown maze must find exit without prior knowledge. It uses search algorithms such as Uniform Cost Search to solve this problem.

Uniform cost Search

- * Expands the node with lowest path cost first.

- * Finds optimal and Least-cost path.

- * Uses a priority queue.

- * Complexity:-

Time and Space-High

Iterative Deepening Search (IDS):-

- * Repeatedly performs depth-limited search with increasing depth.

- * Suitable when goal depth is unknown.

- * Complexity: Low space, ~~high~~ time due to repeated exploration.

Intelligent Agents

- * Simple Reflex Agent: uses only current perception and may get stuck.

* Model-Based Agent:-

Remembers visited paths and may get repetition.

Goal-Based Agent:- Chooses the Safest and Shortest path.

* Utility-Based Agent:- Chooses the Safest and Shortest path.

* A robot is placed in an unknown maze and must find a path to the exit without prior knowledge of the layout.

* Relate the problem to different types of intelligent agent and agent design affects decision making.

Conclusion

UCS is best when path costs vary, while LOS is

Useful when memory is limited. A model-based

Goal Agent with UCS

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